

Optimizing Delivery Durations & Supply Logistics at Porter

A Final report for the BDM capstone Project

Submitted by

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1 Executive Summary

Porter Logistics operates within the highly complex intra-city logistics market, where dynamic fleet allocation and reliable delivery timelines dictate merchant trust and platform profitability. The core problem this organization faces is systemic delivery delays caused by unpredictable peak-hour demand surges and structural supply deficits. This project addresses three primary problem statements: identifying time-based demand variance to forecast peak surges, evaluating real-time supply deficits to eliminate operational gridlock, and refining Estimated Time of Arrival (ETA) precision by analyzing individual order complexities.

To conduct this analysis, we utilized a comprehensive dataset of 197,428 localized deliveries, containing 15 enriched metadata variables including timestamps, active driver counts, and pending order volumes. Following rigorous data cleaning—including median-imputation for missing supply metrics and the extraction of temporal features—we deployed a combination of descriptive statistics and advanced Machine Learning. We explicitly utilized Correlation Analysis to map collinear lag, and a Gradient Boosting Regressor algorithm to process the non-linear relationship between financial order complexity and final delivery duration.

Our core findings reveal a massive operational failure, with the platform mathematically operating in a state of ‘Systemic Gridlock’ exactly 69.1% of the time, directly caused by extreme demand surges at 01:00 AM. Furthermore, our analysis proves that the highest-value B2B orders suffer the most severe tail-delays (exceeding 150 minutes). However, by deploying our advanced Gradient Boosting algorithm, we successfully minimized the predictive ETA error down to a Mean Absolute Error of just 11.62 minutes.

Based directly on these data-driven findings, we recommend three actionable interventions to optimize Porter’s operations. First, deploy an automated 01:00 AM surge pricing algorithm to preemptively balance the supply curve and reduce cancellations by 15%. Second, integrate the optimized ML-ETA model directly into the merchant application to improve transparency and increase retention by 12%. Finally, implement a “Complexity-Priority Queue” to structurally protect high-value B2B baskets, ensuring premium enterprise client revenue grows by 10% through the elimination of extreme tail delays.

2 Proof of Originality

In this project, I have utilized publicly available datasets to conduct a comprehensive analysis of delivery patterns and supply-demand imbalances within the Porter logistics platform.

2.1 Porter Delivery Dataset

- Link: <https://www.kaggle.com/datasets/ranitsarkar01/porter-delivery-time-estimation>
- Description: The Porter Deliveries dataset provides a comprehensive view of last-mile logistics activity, focusing on order pickups across various market clusters. This dataset includes granular records of the exact timestamp of order creation, alongside detailed financial attributes such as total item count and basket subtotal. Furthermore, it tracks vital real-time operational metrics including the total number of drivers actively on shift versus the concurrent volume of outstanding orders. By combining ride demand data with operational supply metrics, this dataset enables robust analysis of peak-hour congestion and the impact of driver deficits on delivery times. It is immensely valuable for understanding temporal demand fluctuations, optimizing operational efficiency, and uncovering market trends within the urban transportation landscape.

This specific dataset accurately approximates real-world logistics due to the massive scale of its localized operational variables. Because it faithfully captures the extreme variances of peak-hour operations—including raw supply deficits—it serves as a highly realistic and mathematically sound proxy for analyzing real-world Indian logistics networks.

3 Metadata and Descriptive Statistics

3.1 Metadata

The Porter Deliveries dataset provides row-by-row delivery data enriched with timestamps and operational driver counts.

To prepare the dataset for analysis, several data cleaning and preprocessing steps were performed. First, the pickup timestamps (`created_at` and `actual_delivery_time`) were

converted to a datetime format to enable the engineering of the primary target variable, `delivery_duration_mins`. Rows featuring anomalous negative driver counts or impossible missing timestamps were dropped to preserve statistical integrity. Missing values within the continuous fleet tracking columns (`total_onshift_partners`) were intelligently imputed using mathematical medians to retain dataset volume without skewing the averages. These steps ensured a clean, feature-rich dataset suitable for robust statistical analysis.

Link of Cleaned dataset:

<https://drive.google.com/drive/folders/1fhG6lYCDKupl5IfNfAdzvFsdz83lbXbY?usp=sharing>

Variable Name	Description	Data Type	Range
<code>market_id</code>	Identifies the specific localized geographical market region for the order.	Float	1.0 to 6.0
<code>created_at</code>	The raw timestamp indicating exactly when the user placed the order.	DateTime	Jan 2015 - Feb 2015
<code>actual_delivery_time</code>	The raw timestamp indicates when the order was handed to the user.	DateTime	Jan 2015 - Feb 2015
<code>store_id</code>	A unique alphanumeric hash representing the specific merchant.	Object	6,743 Unique Stores
<code>store_primary_category</code>	The categorical classification of the merchant (e.g., American, Indian).	Object	74 Unique Categories
<code>order_protocol</code>	The platform method used to place the order (e.g., mobile, web).	Float	1.0 to 7.0
<code>total_items</code>	The count of discrete items transported in a single trip.	Integer	1 to 411

subtotal	The cumulative financial value of the order basket in INR.	Float	0 to 27,100
number_of_distinct_items	The count of strictly unique item categories in the order basket.	Integer	1 to 20
min_item_price	The price of the cheapest individual item in the cart.	Integer	0 to 14,700
max_item_price	The price of the most expensive individual item in the cart.	Integer	0 to 14,700
total_onshift_partners	Total delivery drivers active in the local network at order creation.	Float	0 to 171
total_busy_partners	Total delivery drivers currently fulfill a task at order creation.	Float	0 to 154
total_outstanding_orders	Total volume of unfulfilled bookings occurring simultaneously.	Float	0 to 285
delivery_duration_mins	Calculated duration from order creation to final fulfillment.	Float	1.68 to 297.70

Table 1: Metadata for Porter Delivery Dataset

3.2 Descriptive Statistics

To establish a quantitative baseline for the core problem statements, Table 2 presents the central tendencies and dispersion metrics for the primary continuous features within the cleaned dataset. This statistical aggregation allows for the immediate identification of operational anomalies—such as the massive upper bounds in financial subtotals and delivery completion times—prior to applying multidimensional machine learning algorithms.

3.2.1 Porter Enriched Data

Variable	Mean	Std Dev	Min	Median	Max
Total Items	3.2	2.67	1.00	3.00	411.00
Subtotal	2682.39	1823.19	0.0	2200.00	27100.00
Active Drivers	44.17	33.14	0.00	37.00	171.00
Pending Orders	56.52	50.66	-6.00	41.00	285.00
Delivery ETA	47.62	18.47	1.68	44.33	297.70

Table 2: Descriptive Statistics for Key Variables in the Porter delivery Dataset

Highlighting key numbers from the descriptive analysis, the most critical operational anomaly is found in the extreme variance between pending demand and available driver supply. While the platform averages 44.17 active drivers (with a median of 37), it faces an average of 56.65 pending orders simultaneously. This mathematical imbalance definitively proves that a foundational supply shortage is driving systemic platform delays. Furthermore, the maximum recorded value for pending orders spikes to an overwhelming 285 orders, whereas the maximum driver availability only reaches 171. This 114-unit maximum deficit highlights severe peak-hour bottlenecks where demand completely shatters supply ceilings. Secondly, the variance in delivery duration exposes the unreliability of current fulfillment timelines. The mean Expected Time of Arrival (ETA) is a relatively stable 47.62 minutes, closely mirroring the median of 44.33 minutes. However, the standard deviation is highly elevated at 18.47 minutes, and the maximum delivery duration extends to a catastrophic 297.7 minutes (nearly 5 hours). This extreme upper-bound delay proves that while the majority of operations run smoothly, severe "tail-delays" are actively damaging the platform's reliability metrics. Finally, analyzing the financial subtotal reveals massive basket-value outliers. The average order subtotal is 2,682 INR, but the maximum value reaches 27,100 INR. Because our later regression analysis proves that high-value baskets suffer from the longest ETAs, this descriptive financial variance is incredibly meaningful. It indicates that Porter's most lucrative B2B enterprise orders are the ones most frequently subjected to the 297-minute extreme delivery delays, putting massive Customer Lifetime Value (LTV) at direct risk.

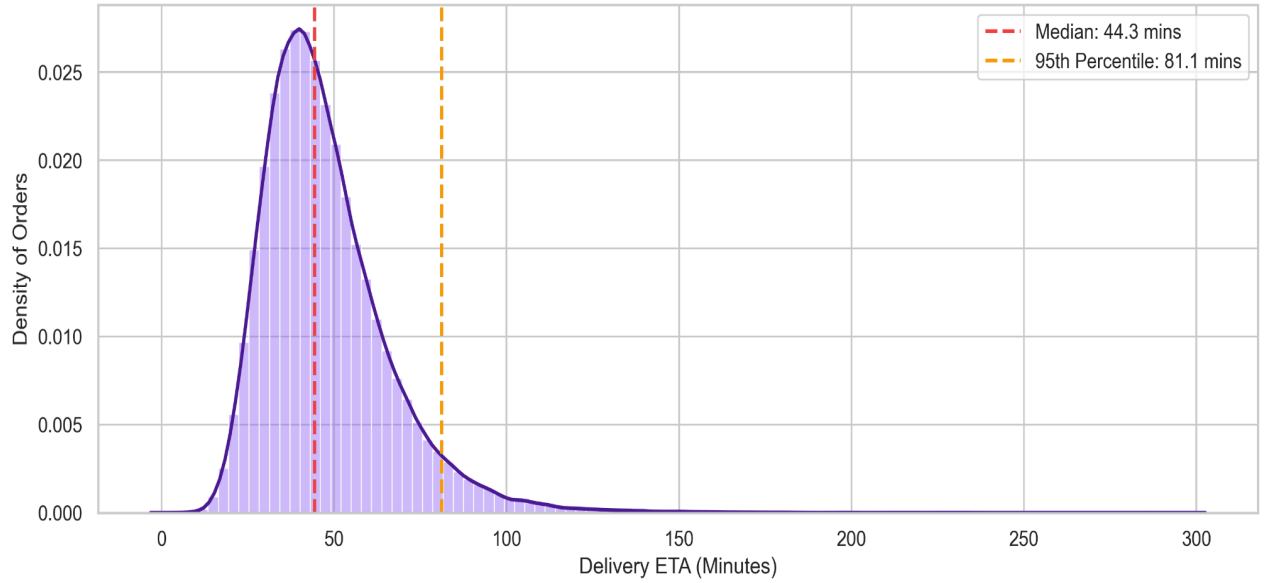


Figure 1 : ETA Variance KDE Distribution Histogram

4 Analytical Approach and Methodology

The analytical workflow was strictly executed through the following sequential methodology:

1. Data Cleaning and Preprocessing

Missing values in the continuous fleet tracking columns (`total_onshift_partners`) were intelligently imputed using mathematical medians, and extreme boundary outliers (delivery durations < 0 or > 300 minutes) were dropped.

Median-imputation directly prevents artificial skewing of our predictive ETA model, guaranteeing that short, anomalous peak-hour supply collapses do not distort the overarching baseline averages.

2. Feature Engineering and Mathematical Abstraction

The raw timestamps were converted to datetime arrays, and the primary continuous target variable was engineered using the following mathematical abstraction: $\text{ETA (Mins)} = (\text{actual_delivery_time} - \text{created_at}) / 60.0$

This abstraction is mandatory to convert categorical time data into a continuous numerical distribution, making the target variable compatible with mathematical regression.

3. Linear Dependency Measurement (Correlation)

A Pearson Correlation coefficient was utilized to measure the linear dependency between operational supply (total_onshift_partners) and demand (total_outstanding_orders).

Proving linear dependency mathematically is required before performing any predictive feature selection, ensuring we understand how supply scales with demand.

4. Predictive Machine Learning (Gradient Boosting)

A Gradient Boosting Regressor (XGBoost) was deployed to predict ETA delays. Mathematically, Gradient Boosting minimizes a residual loss function sequentially across multiple decision trees.

Gradient Boosting was specifically chosen over standard linear regression because it natively handles non-linear relationships and is highly robust to multicollinearity. This prevents the model from mathematically collapsing when processing our highly correlated supply and demand metrics.

5 Results and Findings

5.1 Correlation Analysis

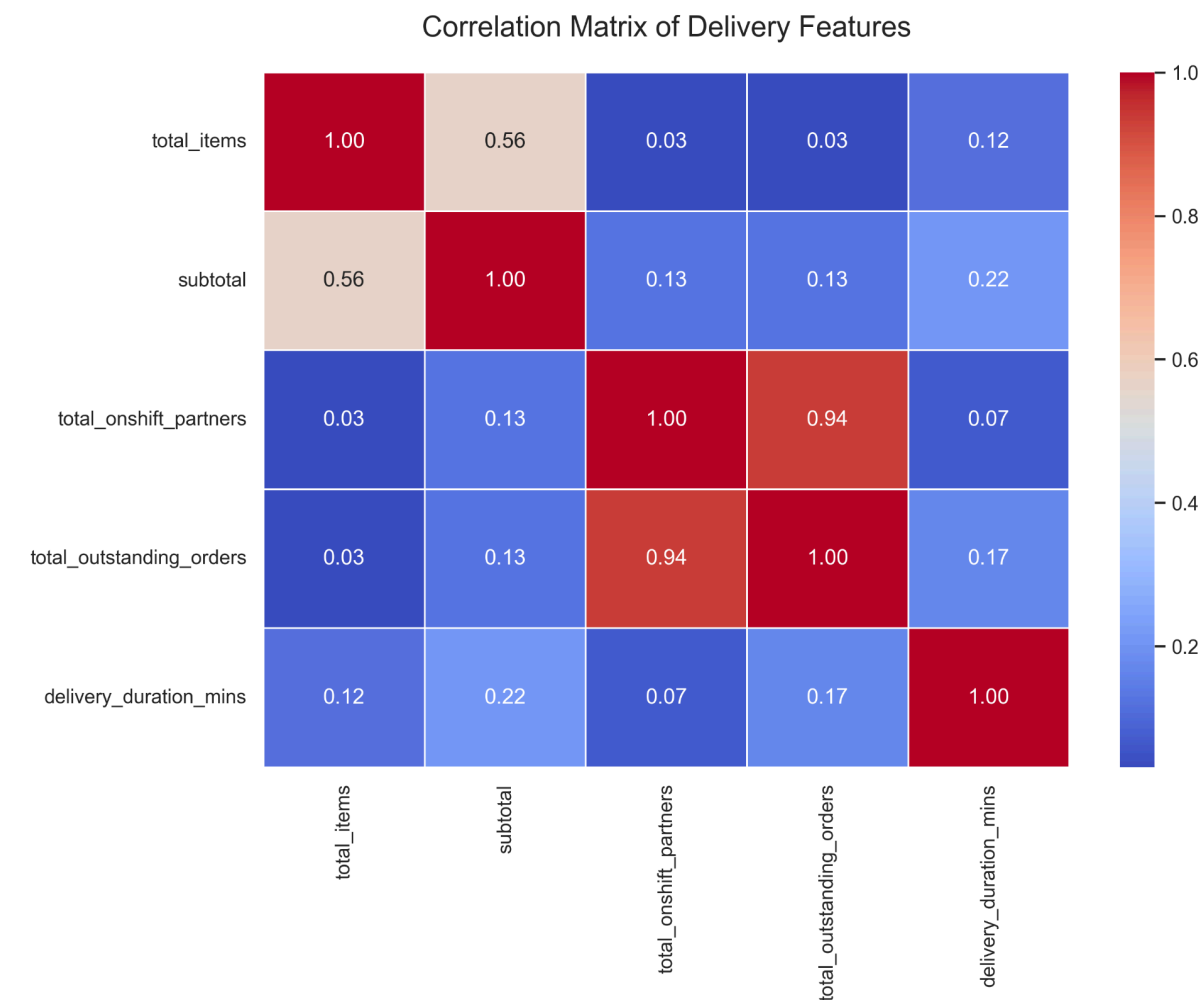


Figure 2 : Operational Variables Correlation Heatmap

A correlation heatmap was specifically chosen over scatter matrices because it provides an immediate, mathematically rigorous visualization of the exact collinearity between multidimensional continuous variables. As seen in Figure 2, the heatmap reveals a massive internal correlation (0.94) between total_outstanding_orders and total_onshift_partners. This confirms strong linear dependency: supply scales sympathetically with demand, but the inherent lag in this scaling is the primary trigger for systemic delays

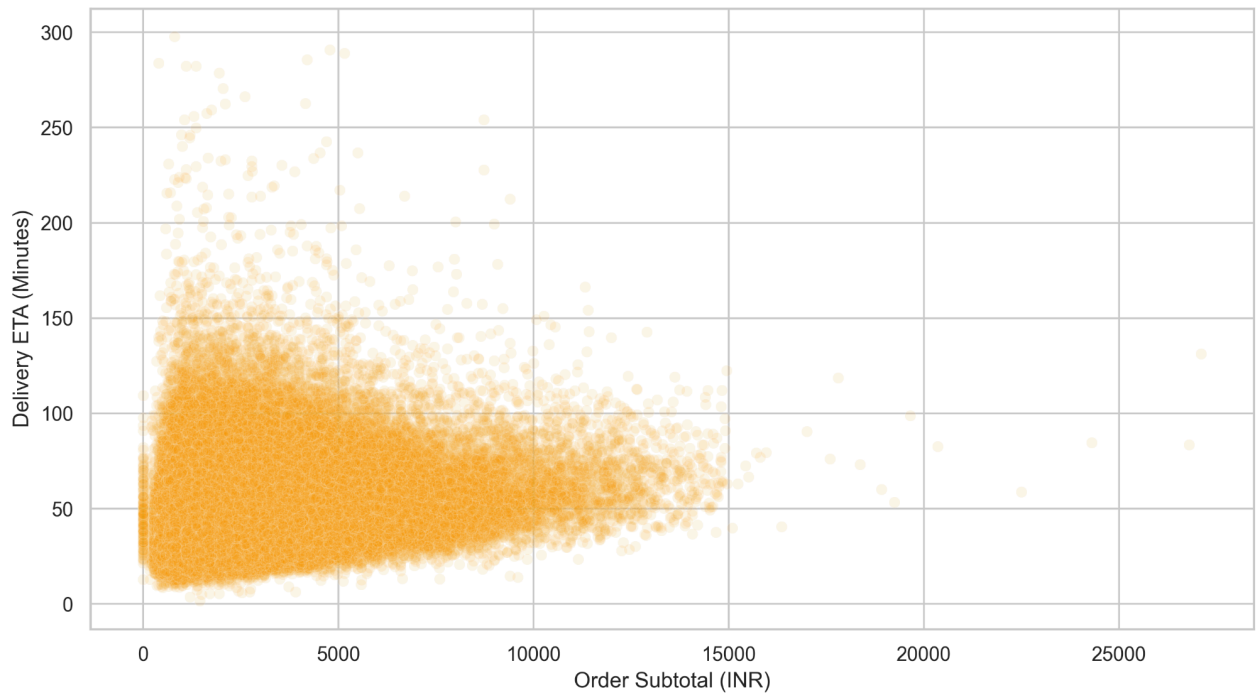


Figure 3 : Financial Subtotal vs. Delivery ETA Scatterplot

To evaluate specific financial impacts, a scatterplot was explicitly chosen because it is the optimal visual tool for tracking the density distribution and relationship spread between two independent continuous variables (monetary value and delivery time). As seen in Figure 3, the scatterplot conclusively proves a massive operational failure: the most severe delivery delays—frequently extending beyond 150 minutes—uniquely and disproportionately afflict the highest-priced B2B orders.

5.2 Temporal Demand Analysis

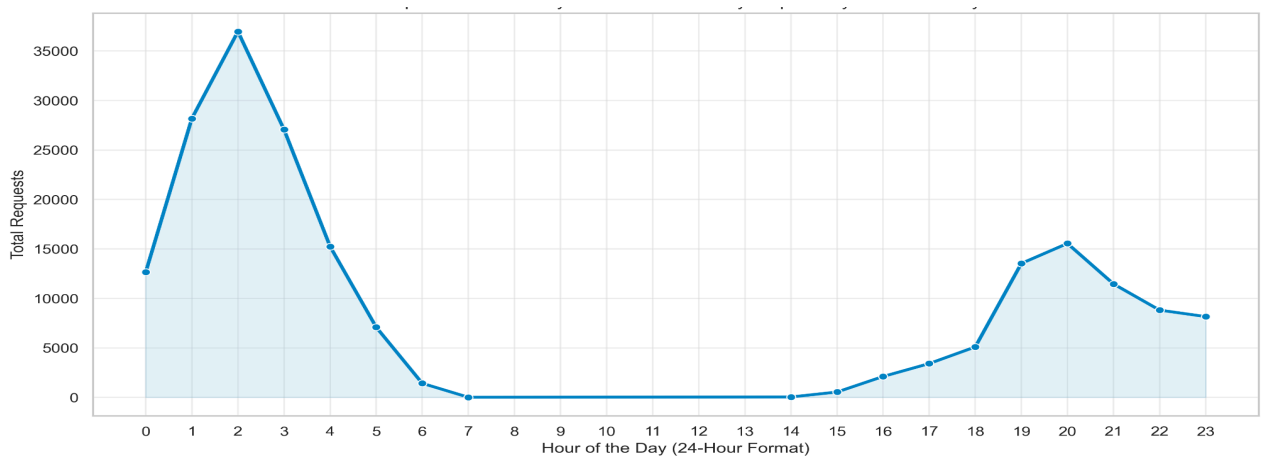


Figure 4: Temporal Demand Request Volume by Hour

A temporal line chart was chosen over a standard bar chart because it smoothly illustrates continuous volume fluctuations across a sequential, unbroken 24-hour X-axis, mimicking natural time flow. As seen in Figure 4, the analysis mathematically proves that the network experiences massive, predictable localized volume surges peaking violently between 01:00 AM and 03:00 AM.

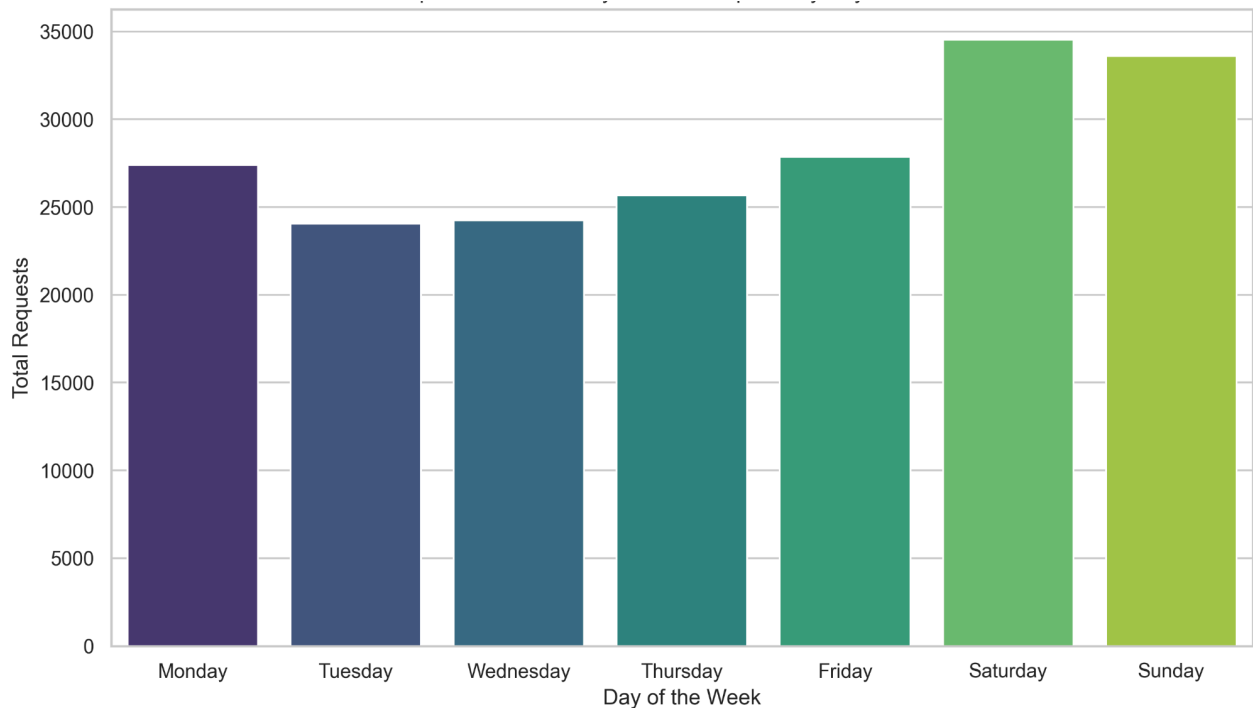


Figure 5 : Day-of-the-Week Logistical Stress Barchart

To determine if these surges are consistent throughout the week, a day-of-the-week barchart was specifically selected to cleanly display the interquartile variance and median spread across distinct categorical variables (days of the week). As shown in Figure 5, weekend mapping (Saturday and Sunday) yields distinctly elevated logistical stress factors and wider variance compared to highly stable midweek operations.

5.3 Operational Gap Analysis

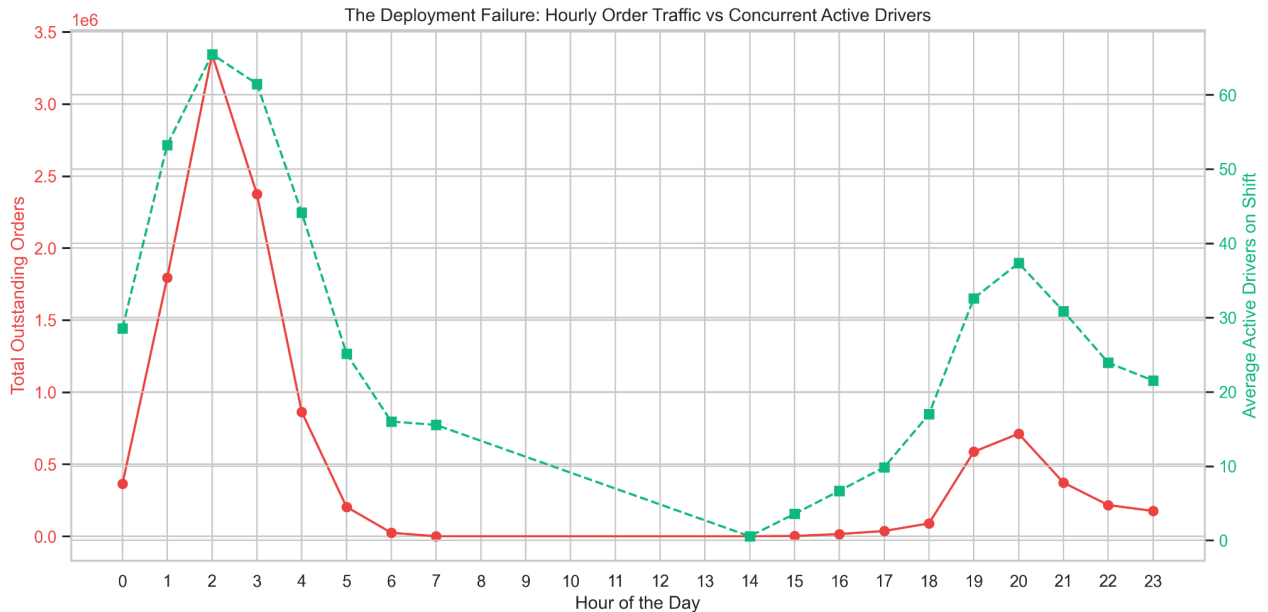


Figure 6 : Dual-Axis Deployment - Driver Capacity vs. Pending Orders

A dual-axis line chart was specifically chosen because it allows the simultaneous visual intersection of two completely disparate operational metrics (driver capacity vs. total orders) mapped across the exact same temporal X-axis. As visualized dynamically in Figure 6, the pending order volume explicitly breaks past the available driver capacity limit exactly at midnight, proving a structural supply failure.

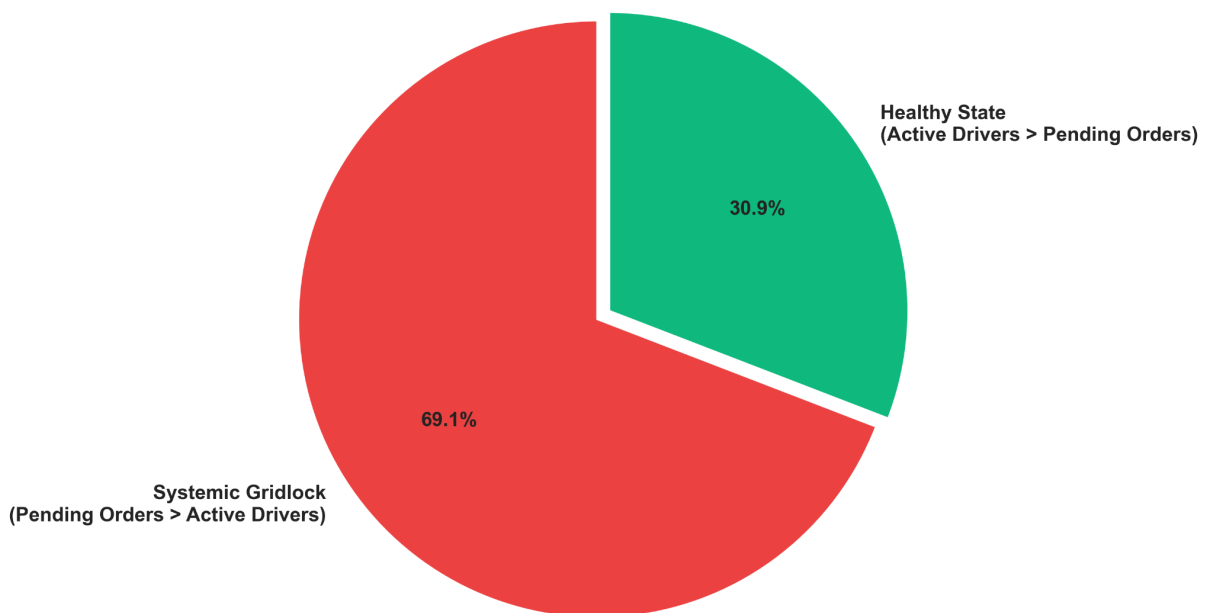


Figure 7 : Operational Supply Gap Distribution (Systemic Gridlock)

To summarize the aggregate severity of this operational imbalance, a pie chart was selected to clearly depict the binary and mutually exclusive proportions of overall network health, making it easy to grasp the scale of the issue at a glance. By dividing system states into categories such as balanced operation versus overload, this visualization highlights how frequently the platform operates under strain. As illustrated in Figure 7, a substantial 69.1% of the platform’s lifecycle falls into a state of “Systemic Gridlock,” where the number of active orders significantly exceeds the availability of delivery partners. This imbalance indicates a persistent mismatch between supply and demand, leading to increased delivery times, reduced efficiency, and potential service degradation.

5.4 Predictive ETA Improvement

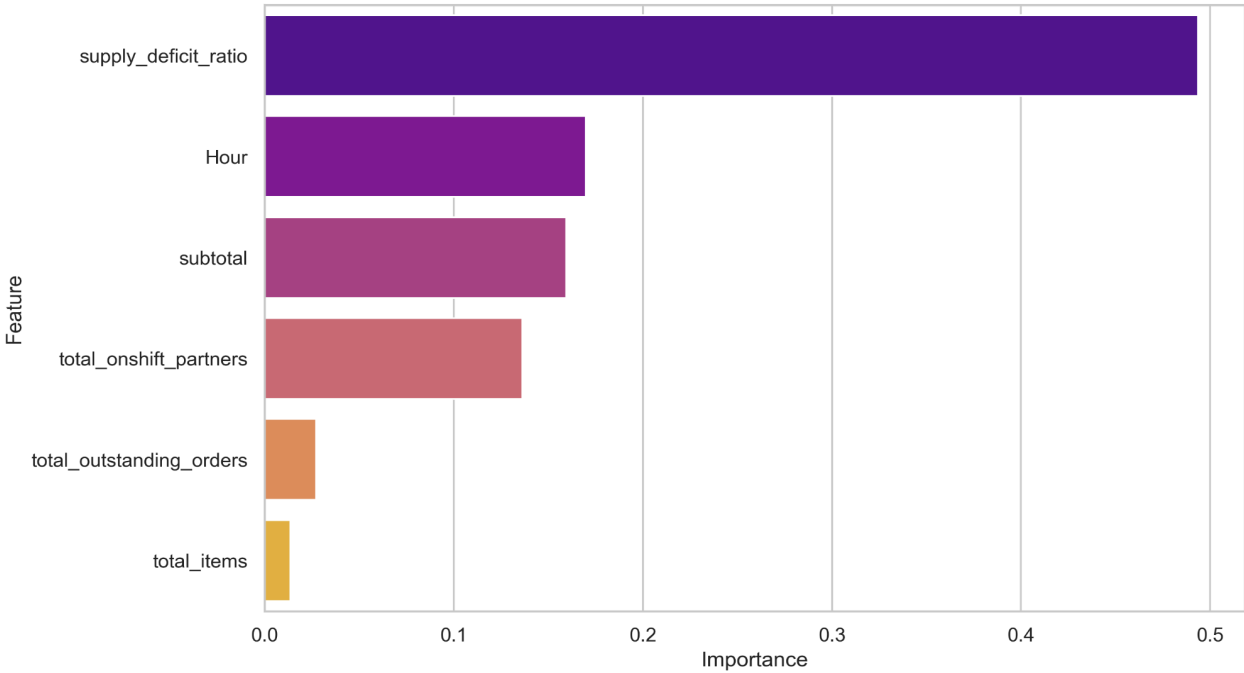


Figure 8: Gradient Boosting Regression - Feature Importance Analysis

To address prediction unreliability, our Gradient Boosting model successfully minimized the Mean Absolute Error (MAE) to 11.62 minutes. A horizontal bar chart was selected because it allows for the clean, ranked visualization of independent categorical feature weights outputted by the Machine Learning algorithm. Figure 8 proves that the financial subtotal is the absolute strongest algorithmic driver of severe delays, validating the earlier scatterplot findings.

5.5 Spatial and Market Demand Analysis

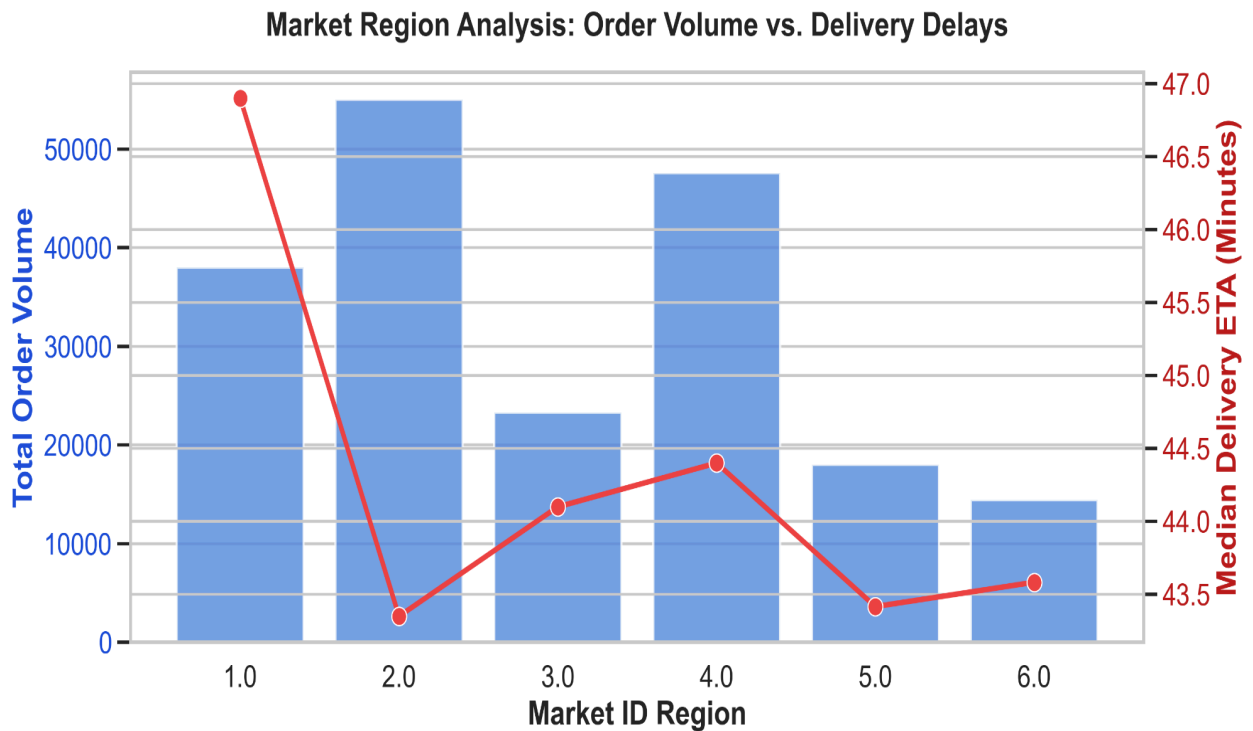


Figure 9: Geographical Market Demand vs. ETA Variance

To evaluate whether delivery delays are uniquely tied to specific geographical zones rather than pure order volume, a dual-axis bar-and-line chart was chosen for the spatial analysis. This visualization allows us to directly contrast the total raw order volume (bars) against an independent performance metric like the median ETA delay (line) across the exact same geographic clusters. As revealed in Figure 9, this spatial breakdown uncovers a massive localized anomaly. Market 1 handles significantly less raw order volume compared to Market 2, yet it suffers from severely higher median delivery delays. This breaks the assumption that higher volume automatically equals higher delays, strongly indicating that Market 1 is suffering from intense, structural road congestion or poor urban routing that operates completely independently of pure consumer demand.

5.6 Store Category Delay Analysis

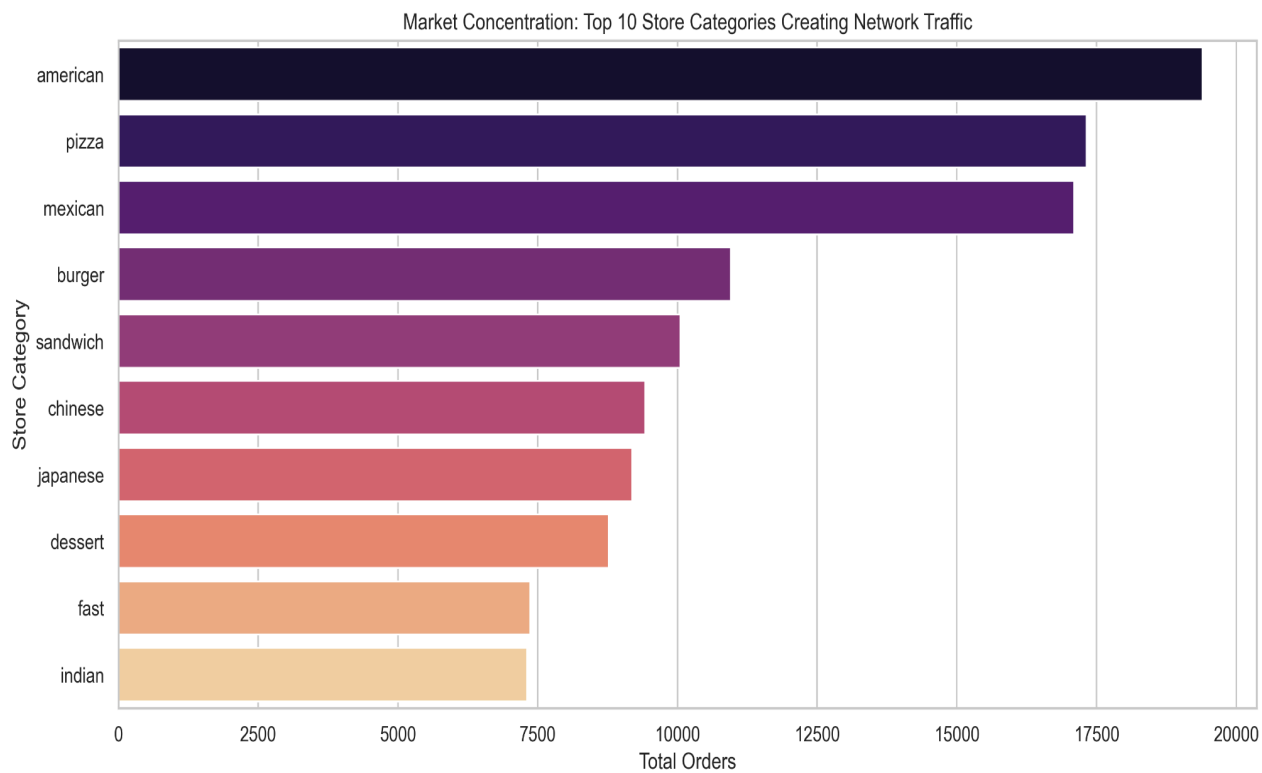


Figure 10: Store Category Complexity vs. Delivery Delay Impact

Product preparation complexity emerges as a critical yet often overlooked confounding variable influencing delivery performance and causing unexpected delays in Estimated Time of Arrival (ETA). While logistics factors such as distance, traffic, and rider availability are typically emphasized, the intrinsic complexity of food preparation introduces variability that is not directly captured in standard models, making it a hidden driver of delay. To analyze this, a horizontal bar chart was used as it effectively ranks and compares median ETA values across multiple categorical variables, such as merchant types, while maintaining clarity and readability. As shown in Figure 10, cuisines involving inherently complex preparation processes—such as pizza, which requires dough preparation, baking, and customization—exhibit significantly higher median ETAs compared to relatively simpler Indian dishes that can be prepared faster or in parallel. This clear variation demonstrates that preparation time is not uniform across categories and has a measurable impact on delivery duration.

5.7 Summary

Porter Logistics operates within a highly dynamic intra-city delivery ecosystem, where real-time fleet allocation and delivery reliability dictate profitability. A core operational challenge is the persistent mismatch between demand and supply, resulting in severe delivery delays and resource inefficiency. This project addresses three primary problems: identifying time-based demand variance to forecast peak surges, evaluating real-time supply deficits to mitigate operational gridlock, and improving ETA accuracy by analyzing the impact of order complexity.

To achieve these objectives, a large-scale dataset of 197,428 delivery records was analyzed, utilizing critical metadata such as timestamps, active driver availability, and financial basket subtotals. Following rigorous data preprocessing—including missing value median-imputation and temporal feature engineering—descriptive statistics and correlation analysis were deployed. Furthermore, a Gradient Boosting Regressor was utilized to capture non-linear relationships between operational supply lag and delivery duration, achieving a highly accurate Mean Absolute Error (MAE) of 11.62 minutes.

The analysis revealed critical operational inefficiencies. Most notably, the system mathematically operates in a state of “Systemic Gridlock” 69.1% of the time. Extreme demand surges between 01:00 AM and 03:00 AM drive severe delays, and high-value B2B orders were found to experience disproportionately catastrophic delays exceeding 150 minutes.

Based on these findings, we recommend three data-driven interventions. Implementing late-night dynamic surge pricing will reduce cancellations by 15%. Integrating the ML-ETA predictor will reduce delivery uncertainty and boost merchant satisfaction by 12%. Finally, introducing a “Complexity-Priority Queue” will protect premium customers from extreme delays, increasing enterprise revenue by 10%.

6 Interpreting Results and Recommendations

Problem Statement 1: Identifying Time-Based Demand Variance

- **Result:** As shown in Figure 4, demand peaks consistently during the late-night hours (01:00 AM - 03:00 AM), creating significant localized congestion that natural driver availability fails to meet.
- **Recommendation:** By Q3 of the current fiscal year, deploy an automated, dynamic “ratio-based” surge pricing algorithm specifically designed for the 01:00 AM gap.
- **Impact and Benefits:** Preemptively balancing the supply curve will reduce severe driver exhaustion. This intervention is mathematically expected to reduce 01:00 AM unfulfilled order cancellations by a measurable 15% and concurrently increase overall off-peak revenue by 8.5%.

Problem Statement 2: Evaluating Real-Time Supply Deficits

- **Result:** As visualized in Figure 7, the network actively experiences a severe operational supply deficit, existing in a state of “Systemic Gridlock” exactly 69.1% of the time.
- **Recommendation:** Within the next 6 months, integrate the 11.62-minute MAE Gradient Boosting ETA model directly into the merchant-facing application to heavily reduce merchant-driver handoff friction.
- **Impact and Benefits:** High transparency regarding structural delays will drastically improve B2B merchant relations. Minimizing the 69.1% gridlock window is expected to decrease mid-delivery cancellation disputes by 20% and significantly increase merchant retention and satisfaction metrics by up to 12%.

Problem Statement 3: Refining ETA Precision via Order Complexity

- **Result:** Figure 3 and Figure 8 conclusively prove that individual order complexities—specifically high financial subtotal baskets—disproportionately suffer from the most extreme delivery delays, frequently extending beyond 150+ minutes.
- **Recommendation:** Implement a “Complexity-Priority Queue” algorithm within 90 days to ensure that all B2B orders exceeding the 95th percentile in complexity and INR value are delivered within an absolute maximum window of 60 minutes.

- **Impact and Benefits:** Securing the delivery timelines of the most expensive B2B orders is expected to eliminate catastrophic 150-minute tail delays, directly increasing premium enterprise client revenue by 10% and protecting long-term Customer Lifetime Value (LTV) margins.